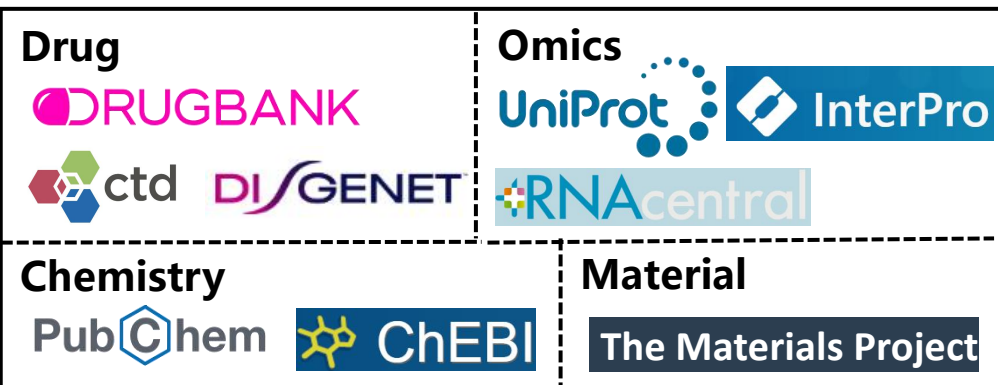


a. Data Acquisition

01 Structured databases



02 Unstructured text



Publications

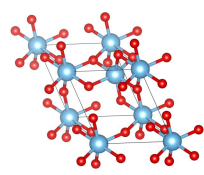
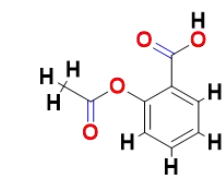
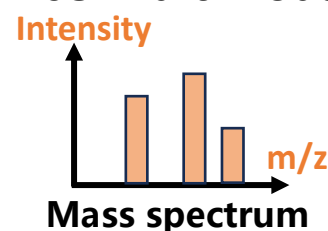


Patents



Scientific reports

03 Multimodal data



b. Knowledge Extraction

01 Rule/dictionary-based extraction

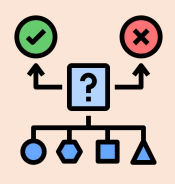
Step1: Prepare Lexicon and rules

Domain-specific dictionary



e.g., "aspirin"

Pattern rules



e.g., "X inhibits Y"

Step2: Match and Extract



Match terms & patterns

(Drug, inhibits, Protein)

02 LLM-based extraction

Step1: Fine-tuning



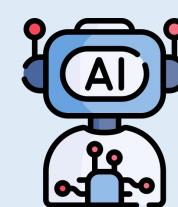
Foundational LLM



Domain-specific LLM

Scientific datasets

Step2: Prompt engineering



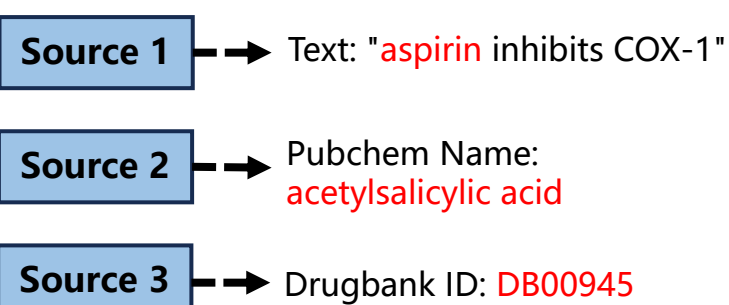
Prompt instruction

"Extract chemical entities (e.g., compounds, enzymes) and their interaction relationships from the text below."

LLM-based: automatically learns complex patterns from data ✓

c. Integration and Representation

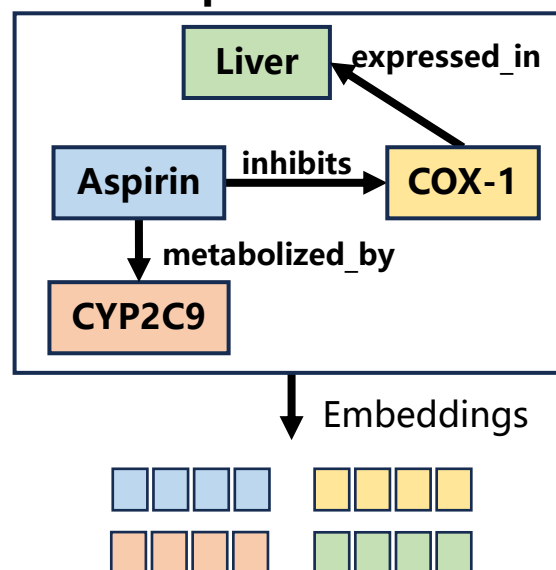
01 Integration & Alignment



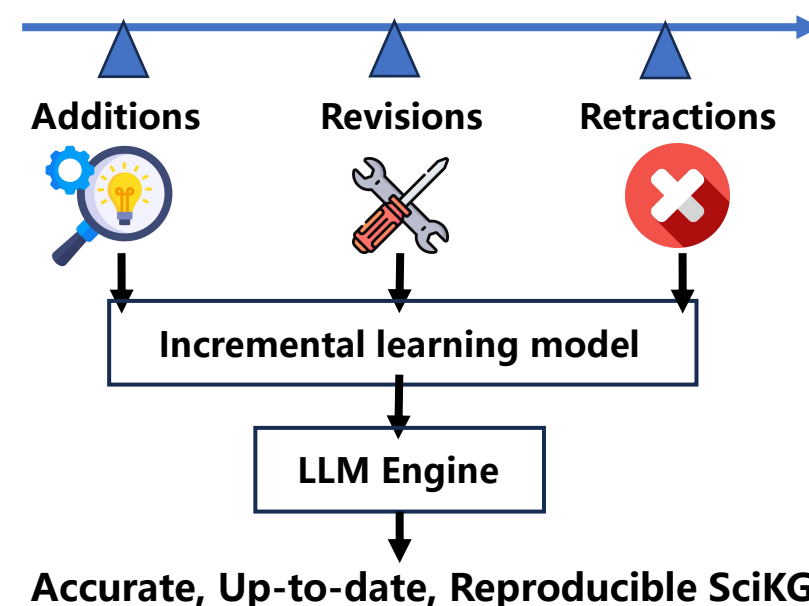
Map to unified ontology

CHEBI: 15365

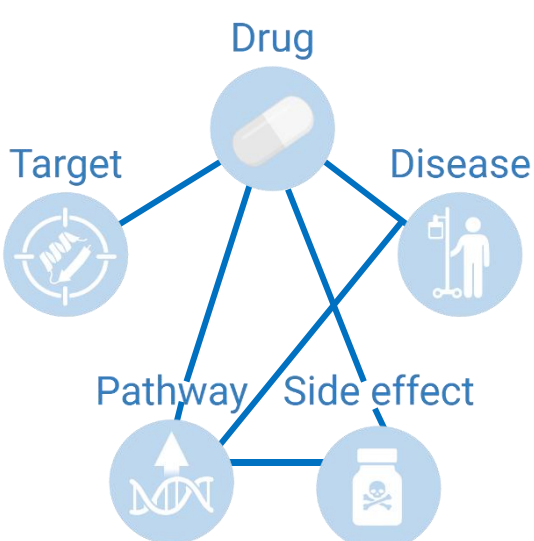
02 Representation



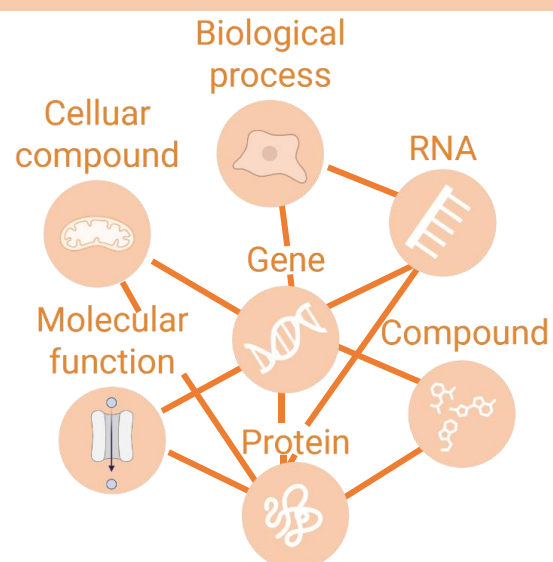
d. Dynamic Updating and Maintenance



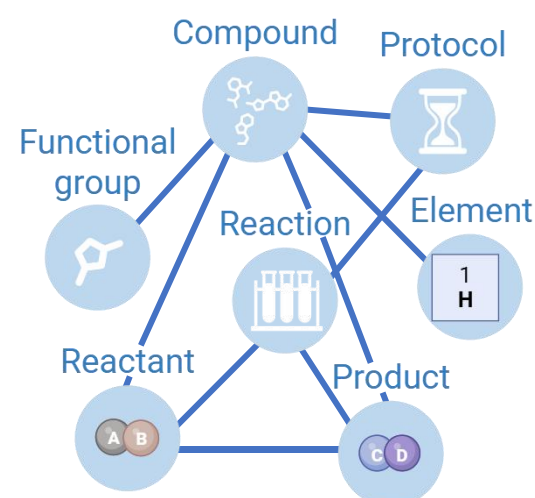
e. Drug KG



f. Biomolecular KG



g. Reaction KG



h. Material KG

